

Multiple Choice

1. **Answer:** A

Options: A) carnitine.; B) pyruvate.; C) NADH.; D) glycerol.; E) thiokinase.

Note: Fatty acids must be transported into the mitochondria as acylcarnitine complexes because the inner mitochondrial membrane is impermeable to free fatty acids, and carnitine facilitates their transfer across this barrier for subsequent beta-oxidation.

2. **Answer:** A

Options: A) It is always associated with psoriasis; B) It is caused by deposition of uric acid crystals in the joints; C) It is an example of an oligoarthropathy; D) Heberden's nodes are a feature; E) Rheumatoid factor is present in 25% of rheumatoid cases

Note: The statement is incorrect; rheumatoid arthritis is not always associated with psoriasis, as they are distinct conditions, and while they can co-occur, one does not necessarily imply the presence of the other.

3. **Answer:** B

Options: A) Watson; B) Smith; C) Pavlov; D) Whitebear; E) Johnson

Note: The correct answer is Smith because she is recognized for her longitudinal research on cognitive aging, which demonstrated that retrieval difficulty tends to increase with age.

4. **Answer:** A

Options: A) Enhance visual acuity; B) Promote weight gain; C) Increase memory and intelligence; D) Produce emotional stability; E) Reduce allergic reactions

Note: The activation of sirtuins has been linked to the regulation of cellular processes that may improve mitochondrial function and protect retinal cells, thereby enhancing visual acuity.

5. **Answer:** A

Options: A) 1 in 100; B) 1 in 50; C) 1 in 5; D) 1 in 10; E) 1 in 1000

Note: The correct answer is 1 in 100 because epidemiological studies indicate that approximately 1% of newborns are born with detectable congenital abnormalities, making this a widely accepted statistic in health sciences.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The afferent limb of the pupillary light reflex is true because it involves the transmission of light stimulus information through the optic nerve, which carries visual signals from the retina to the brain.

7. **Answer:** True

Options: A) True; B) False

Note: AIDS is not among the top 10 causes of death in the U.S. primarily because of advances in antiretroviral therapy, which have significantly reduced mortality rates associated with the disease, leading to a decline in its prevalence as a leading cause of death.

8. **Answer:** True

Options: A) True; B) False

Note: Sleep disorders are not specifically included as diagnostic criteria for Anorexia Nervosa according to the DSM-5, which focuses on eating behaviors, body image disturbances, and weight-related issues rather than sleep-related symptoms.

9. **Answer:** False

Options: A) True; B) False

Note: Excessive consumption of fast food is not a significant cause of iodine deficiency because fast food is typically low in essential nutrients, including iodine, rather than high in nutrient content.

10. **Answer:** True

Options: A) True; B) False

Note: A₂L catheter bag is recommended for overnight use because it provides sufficient capacity to safely accommodate increased urine output during the night, reducing the risk of overflow and maintaining patient comfort and hygiene.

Long Form Response

11. **Answer:** Age serves as a critical variable in gerontological research, influencing both health outcomes and the effectiveness of interventions targeted at the elderly population. As individuals age, they experience a range of physiological and psychological changes that can affect their health status, including increased susceptibility to chronic diseases, cognitive decline, and alterations in social support systems. These age-related factors necessitate tailored interventions that consider the unique needs of older adults, such as the implementation of age-specific health promotion strategies and the adjustment of treatment protocols to accommodate varying levels of resilience and comorbidities. Furthermore, understanding the significance of age allows researchers and practitioners to identify trends and disparities in health outcomes, thereby fostering the development of targeted policies and programs aimed at improving the quality of life for older individuals. Overall, age is a fundamental determinant in gerontological studies, shaping both the context and content of health-related interventions for the aging population.

Note: correct because it emphasizes how age-related physiological and psychological changes directly impact health outcomes and the need for age-specific interventions in gerontological research.

12. **Answer:** An unfavorable fracture of the mandible can significantly alter the biomechanical dynamics of the jaw, particularly impacting the position and function of the anterior fragment. The lateral pterygoid muscle, which is primarily responsible for the depression and protraction of the mandible, plays a crucial role in this displacement. When a fracture occurs, the contraction of the lateral pterygoid on the fractured side can lead to the anterior fragment being displaced anteriorly. This anterior displacement can compromise occlusion, affect the functionality of the masticatory system, and result in asymmetrical jaw movement. Understanding the influence of the lateral pterygoid muscle on fracture displacement is essential for developing effective treatment plans and ensuring optimal recovery of mandibular function.

Note: correct because it accurately describes how the lateral pterygoid muscle's function in mandible movement can lead to the anterior fragment being displaced anteriorly during an unfavorable fracture, thus illustrating the biomechanical implications of such injuries.